COMPUTER SCIENCE – LEVEL 6

BASIC ELECTRONICS

CT/CU/CS/CC/01/6/8

1 ½ HOURS

AUGUST 2025

Faradays

JKUAT TVET INSTITTUTE – KAREN CAMPUS

WRITTEN ASSESSMENT

CAT 2

R - opposition electric energy in

Capacitor - store

an electric field.

Answer ALL questions

SECTION A (20 Marks) R(Ω)

Each question contains (4 marks)

1. Define an electric circuit and list THREE basic components required for a complete circuit. (4 marks)

2. Differentiate between resistors and capacitors in terms of:

   i). Function in a circuit (1 mark)

   ii). Unit of measurement R (ohms) C (Farads) (1 mark)

   iii). Symbol representation R(Ω) C (F) (1 mark)

   iv). Energy storage mechanism R(Oppose) C(Electric Field) (1 mark)

3. With the aid of a labeled diagram, explain the operation of a PN junction diode under forward bias. (4 marks)

4. Classify computer memory into TWO main categories and provide TWO examples of each. (4 marks)

   i) Cachemem - Secondary - Optical Disk

   ii) Secon - Primary - RAM, Rom

5. Outline FOUR differences between N-type and P-type semiconductors. (4 marks)

6. Convert the following numbers as indicated:

   i). Binary 1101 to decimal (2 marks)

   ii). Decimal 25 to binary (2 marks)

N	P
i) Doped with Pentavalent	- Trivalent
ii) Donor atom	- Acceptor atom
iii) Electron	- Few electrons
iv) Few holes	- Many holes

SECTION B (30 Marks)

7. a) Define the term "semiconductor" and explain why silicon is commonly used in electronic devices. (3 marks)

b) Compare intrinsic and extrinsic semiconductors using the following criteria:

i). Definition (2 marks)

ii). Conductivity (2 marks)

iii). Charge carriers (2 marks)

iv). Doping process (2 marks)

c) Electronic components can be classified into Active, Passive and Electromechanical components. Describe each classification and in each case give TWO examples. (9 marks)

8. Discuss FIVE emerging trends in electronic manufacturing and highlight ONE challenge associated with each trend. (10 marks)

i) Cyber Crime
ii) Skilled Workers
iii) E - waste
iv) Pollution
v) Internet of Things

Active
Passive
Active
i) They amplify
ii)
Passive
Do not Amplify
Cannot generate power

CAT 2

42/50

JKUAT CONTINOUS ASSESSEMENT TESTS

NAME: WAHIRA JOSHUA NYAGA IVET

REG NO: TDC113-C002-0003/2024

COURSE: Computer Science YEAR OF STUDY: 1

SHEET NO:

UNIT CODE: TITLE: Basic Electronics

DATE: 9/5/2025

NOTE: This stationery will be used for Continuous Assessment work only.

It will be a breach of examination regulations to use it otherwise

SECTION A

1. Electronic circuit refers to a loop or pathway where current flows through. Components required i) Source ii) Wire iii) Switch

(4)

2. i) Resistors they usually resist the flow of ~~electrons the~~ electric current while capacitors they usually store energy in the electric field.

ii) Resistors are measured in ohms (Ω) while Capacitors are measured in Farads (F)

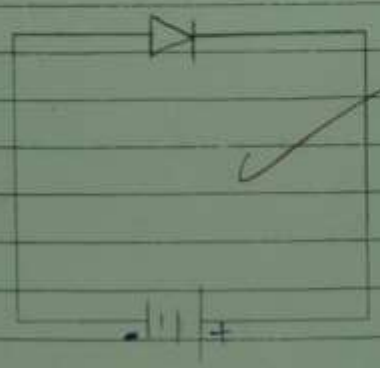
iii) ~~Resistor → Ω~~

iii) Resistor is represented with the symbol (Ω) R while Capacitor is represented with the symbol (F) C

iv) Capacitor stores their energy in the electric field while resistors. They regulate the amount of current so as to avoid expense of energy.

(03)

3)



(2)

09

4)

i) Volatile memory ~~These~~ This a memory which stores data temporary and can be lost easily when the power is switched off

4) i) Secondary storage refers to non-volatile memory which stores data for future reference eg Blue Ray Disk and Flash memory

ii) Main Memory Refers to memories which store data either permanently or temporary eg RAM and ROM

5) i) N-type are said to be pentavalent elements while P-type are said to be tri-valent elements

ii) N-type are said to be the donor atoms while P-type are said to be the acceptor atoms

iii) N-type, electrons are the major charge carriers while P-type, electrons are minor charge carriers

iv) N-type, holes are minor charge carriers while P-type, holes are major charge carriers

6) i) 1101_2 to decimal

$(2^3 \times 1) + (2^2 \times 1) + (2^1 \times 0) + (2^0 \times 1)$

$8 + 4 + 0 + 1 = 13$

$= 13_{10}$

2

ii) 25_{10} to binary

2	25
2	12 R 1
2	6 R 0
2	3 R 0
2	1 R 1
	0 R 1

$25_{10} = 11001_2$

2

08

SECTION B

7a) Semiconductors refers to materials which partially conducts electricity with the properties between conductors and insulators (1)

b) i) Intrinsic are pure semiconductors which have good electrical conductivity while extrinsic are impure semiconductors which are added impurities known as the dopants to conduct electricity.

ii) Extrinsic semiconductors are said to have better electrical conductivity more than intrinsic semiconductors because they have high forbidden energy in between the conduction band and the valence band. (08)

iii) Extrinsic semiconductors electrons and holes are their charge carries while intrinsic delocalised electrons are their charge carriers.

iv) Extrinsic semiconductors they have to undergo through doping process for them to pure while intrinsic semiconductors are already pure so they don't need to undergo doping process

c) Active components are components which require an external power source for them to work an example is a Transistor which acts as an amplifier and also diodes while passive components they don't require the external power source for them to work in fact they distipate and store energy an example is a i) Capacitor
ii) Resistance (06)

8 i) Skilled workers in Technology - Since the world is evolving and developing technology, getting highly skilled workers in the industry to operate the machines is a major challenge.

ii) E-waste - Since the many industries are developing various features like machines and other products which uses harmful chemicals like plastics they do produce harmful poisonous gas which tends to affect the O2 analogy (08)

iii) Artificial Intelligence - Since AI technology has made work easier and become common all over the world still it also brings a challenge job replacement and displacement

iv) IOT - Internet of things being one of the most relied technology by many people, it has also brought luxurious life to many people they rely on others to work and fustion to them. (3)

x) Raw materials - Since raw materials are the main ~~p~~ ingredients used by the industry to generate and make ~~the~~ products, they are also becoming expensive ~~for them~~ with rising cost hence the living standard is high.

02

02